Numerical Simulation of Financial SDEs

Problem 4: GBM Benchmark and Non-Affine Volatility

Team #6

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Numerical Partial Differential Equations

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Abstract

This report compares exact simulation, Euler–Maruyama (EM), and scalar Milstein for geometric Brownian motion (GBM), then studies a non-affine stochastic-volatility model without an exact transition sampler. All resolution comparisons share Brownian paths through aggregation of fine increments. In GBM, the measured terminal strong slopes are 0.500 for EM and 0.991 for Milstein, consistent with orders one half and one. Both methods nevertheless have identical first-order bias in the terminal mean. Paired estimates, independently fitted control variates, and an analytic discrete-mean identity separate this bias from sampling noise. Histograms, quantiles, moments, and positivity diagnostics supply additional checks. For the prescribed non-affine model, simultaneous log-price EM produces positive prices, and three nested numerical references yield an empirical strong slope of 0.491 on the stated fitting window. Fine-grid call-payoff differences have confidence intervals containing zero, so no resolved weak order is claimed. Independent-seed repetitions assess variability of fitted rates, while a measured kernel-cost study compares GBM methods at a common accuracy threshold. Numerical-reference agreement and sampling intervals remain evidence rather than exact-error bounds.

Keywords: SDE; Euler–Maruyama; Milstein; Monte Carlo; nested grids.

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1 Introduction and motivation

Asset prices are driven by both systematic growth and random shocks. Numerical SDE methods must therefore approximate random paths and expectations, rather than only one deterministic trajectory. We chose this direction because GBM provides an exact benchmark for checking an implementation, whereas non-affine volatility requires a numerical reference. The contrast exposes what a convergence plot can establish and what remains uncertain [1, 2, 3].

1.1 Interpreting the assignment

The first page of the supplied problem pack explicitly specialises the generic ODE project brief for Direction 4. Exact–EM–Milstein comparison, strong and weak convergence with coupled paths, Monte Carlo uncertainty, and non-affine nested-grid validation replace deterministic stability-region and adaptive-step analyses. Consequently RK4, implicit-Euler Newton iterations, stiffness ratios, and an adaptive ODE controller are not additional tasks here. We retain the supplied report template's narrative structure, pseudocode, verification, cost comparison, references and transparency appendices.

1.2 GBM model and exact oracle

With initial price $S_0 > 0$, drift μ, constant volatility $\sigma > 0$, and standard Brownian motion W, the model is

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t, \qquad 0 \le t \le T. \tag{1}$$

Time is measured in years and prices in currency units; μ has units year^{-1} and σ year$^{-1/2}$. Itô's lemma, with $f'(S) = 1/S$ and $f''(S) = -1/S^2$, gives

$$dX_t = (\mu - \tfrac{1}{2}\sigma^2)\,dt + \sigma\,dW_t, \qquad X_t = \log S_t. \tag{2}$$

The exact pathwise oracle and terminal moments are

$$S_t = S_0 \exp\{(\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\}, \tag{3}$$

$$m = \mathbb{E}[S_T] = S_0 e^{\mu T}, \tag{4}$$

$$v = \mathrm{Var}(S_T) = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1), \tag{5}$$

$$q_p = S_0 \exp\{(\mu - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\Phi^{-1}(p)\}. \tag{6}$$

We selected $S_0 = 100$, $\mu = 0.05$, $\sigma = 0.30$, $T = 1$ for GBM; the pack does not prescribe these GBM values. Equations (4)–(5) give mean 105.1271 and variance 1040.7868. An exact terminal value is calculated using the same W_T as the numerical path, not a separate random draw.

1.3 Prescribed non-affine model

With $X_t = \log S_t$, the second model is

$$dX_t = [\mu - \tfrac{1}{2}g(Y_t)^2]\,dt + g(Y_t)\,dW_t^{(1)}, \tag{7}$$

$$dY_t = \kappa(\theta - Y_t)\,dt + \xi\sqrt{1 + Y_t^2}\,dW_t^{(2)}, \tag{8}$$

$$g(y) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}}, \qquad d\langle W^{(1)}, W^{(2)}\rangle_t = \rho\,dt. \tag{9}$$

Here Y is a dimensionless latent volatility factor, $g(Y)$ is instantaneous asset volatility, κ is a mean-reversion rate, θ its long-run factor level, ξ the factor's noise scale, and ρ the instantaneous driver correlation. The bounded logistic function keeps g between two positive limits. The nonlinear factor diffusion prevents using the GBM exact transition formula.

Table 1: The prescribed non-affine baseline, reproduced without changing the pack's parameters.

S_0	Y_0	μ	κ	θ	ξ
100	0	0.05	2	-0.2	0.6
ρ	$\sigma_{\min}$	$\sigma_{\max}$	T	X_0	strike K
-0.7	0.10	0.50	1	$\log 100$	100

Sections 2 and 3 explain the methods and implementation. Section 4 compares accuracy, uncertainty and cost. Section 5 states the limitations.

2 Discretisation, convergence and stochastic stability

2.1 Three GBM updates

Let $t_n = nh$, $h = T/N$ and $\Delta W_n = W_{t_{n+1}} - W_{t_n}$. Brownian motion has independent increments with

$$\Delta W_n = \sqrt{h} Z_n, \quad Z_n \sim \mathcal{N}(0,1), \quad \mathbb{E}[\Delta W_n] = 0, \quad \mathbb{E}[(\Delta W_n)^2] = h. \tag{10}$$

The variance is h, whereas the standard deviation is $\sqrt{h}$. Exact simulation, price-based EM and scalar Milstein are respectively

$$S_{n+1}^{\text{exact}} = S_n^{\text{exact}} \exp\{(\mu - \tfrac{1}{2}\sigma^2)h + \sigma \Delta W_n\}, \tag{11}$$

$$S_{n+1}^{E} = S_n^{E}(1 + \mu h + \sigma \Delta W_n), \tag{12}$$

$$S_{n+1}^{M} = S_n^{M}[1 + \mu h + \sigma \Delta W_n + \tfrac{1}{2}\sigma^2((\Delta W_n)^2 - h)]. \tag{13}$$

An update formula advances an approximation by one time step from known old values. Milstein includes $\frac{1}{2} b(S) b'(S)((\Delta W)^2 - h)$, with $b(S) = \sigma S$ [2, 3]. EM on the GBM log price instead reproduces equation (11) exactly; its Milstein correction vanishes. It is therefore inappropriate for demonstrating a nonzero GBM time-discretisation error.

2.2 Strong versus weak error and convergence order

The terminal errors used throughout are

$$e_s(h) = \mathbb{E}|S_N^{(h)} - S_T|, \tag{14}$$

$$e_w(h; \varphi) = |\mathbb{E}[\varphi(S_N^{(h)})] - \mathbb{E}[\varphi(S_T)]|. \tag{15}$$

Strong error compares corresponding realizations; weak error compares an observable's expectation. Neither is a maximum-in-time path error. If $e(h) \sim Ch^p$ with $C > 0$, then $\log e(h) \sim \log C + p \log h$, so the log–log slope approaches p.

For GBM, expanding the exact multiplier shows the leading term that EM omits:

$$e^{(\mu - \sigma^2/2)h + \sigma \Delta W} = 1 + \mu h + \sigma \Delta W + \tfrac{1}{2}\sigma^2[(\Delta W)^2 - h] + O_{L^2}(h^{3/2}). \tag{16}$$

The centered quadratic increment has variance $2h^2$. The martingale part of the local error has centered, orthogonal increments across steps: its scale accumulates like a square root of the number of steps, rather than their absolute sum, giving $\sqrt{T/h}\, h = O(h^{1/2})$. The state-weighted errors need not be independent. Milstein cancels that leading term; its remaining local L^2 defect is $O(h^{3/2})$, with local mean defect $O(h^2)$, yielding a global $O(h)$ scale. This is an explanatory order argument, not a complete proof. The standard strong-convergence theorem, using globally Lipschitz coefficients and the required coefficient derivatives and finite moments, justifies EM order $1/2$ and scalar Milstein order 1 in this GBM setting [2, 3]. It also implies the corresponding terminal L^1 upper bounds by Cauchy–Schwarz.

For the particular weak observable $\varphi(S) = S$, both schemes have the same conditional mean because the noise terms in equation (10) are centered:

$$\mathbb{E}[S_{n+1}^{E} \mid \mathcal{F}_{t_n}] = (1 + \mu h) S_n^{E}, \tag{17}$$

$$\mathbb{E}[S_{n+1}^{M} \mid \mathcal{F}_{t_n}] = (1 + \mu h) S_n^{M}, \tag{18}$$

$$\mathbb{E}[S_N^{E}] = \mathbb{E}[S_N^{M}] = S_0(1 + \mu h)^{T/h}. \tag{19}$$

Using $\log(1 + \mu h) = \mu h - \mu^2 h^2/2 + O(h^3)$ gives the signed bias

$$b_{\text{exact}}(h) = S_0(1 + \mu h)^{T/h} - S_0 e^{\mu T} = -\tfrac{1}{2} S_0 e^{\mu T} \mu^2 T h + O(h^2). \tag{20}$$

The nonzero leading constant for our $\mu = 0.05$ proves first-order mean bias. If $\mu = 0$, this mean bias vanishes and there is no slope to measure. Identical mean bias does not imply identical biases for every payoff.

2.3 Non-affine log-EM and correlated noise

From independent $Z_{1,n}, Z_{2,n} \sim \mathcal{N}(0,1)$, generate

$$\Delta W_n^{(1)} = \sqrt{h} Z_{1,n}, \qquad \Delta W_n^{(2)} = \sqrt{h}(\rho Z_{1,n} + \sqrt{1-\rho^2} Z_{2,n}). \tag{21}$$

Both increments have variance h, and their covariance is ρh. The simultaneous EM update is

$$X_{n+1} = X_n + [\mu - \tfrac{1}{2} g(Y_n)^2]h + g(Y_n)\Delta W_n^{(1)}, \tag{22}$$

$$Y_{n+1} = Y_n + \kappa(\theta - Y_n)h + \xi\sqrt{1+Y_n^2}\Delta W_n^{(2)}, \tag{23}$$

$$S_{n+1} = e^{X_{n+1}}. \tag{24}$$

Both right-hand sides use Y_n, not the newly computed Y_{n+1}. We do not implement multidimensional Milstein. With $a(y) = \xi\sqrt{1+y^2}$ and $c = \sqrt{1-\rho^2}$, the diffusion columns relative to independent drivers and their commutator are

$$b_1 = (g, \rho a)^\top, \quad b_2 = (0, ca)^\top, \quad (Db_2)b_1 - (Db_1)b_2 = (-ca\, g'(y), 0)^\top \neq 0. \tag{25}$$

Cross terms and iterated stochastic integrals cannot in general be removed. Copying the scalar GBM correction would not justify strong order one [3].

2.4 Moment checks and positivity instead of ODE stability plots

For GBM's price methods, the first and second expected multiplier moments are

$$\mathbb{E}[A] = 1 + \mu h, \qquad \mathbb{E}[A_E^2] = (1+\mu h)^2 + \sigma^2 h, \qquad \mathbb{E}[A_M^2] = \mathbb{E}[A_E^2] + \tfrac{1}{2}\sigma^4 h^2. \tag{26}$$

They give finite discrete terminal moments and independent analytic checks. Growth of moments is not itself instability: the physical GBM mean and variance also grow. We do not claim unrestricted numerical stability from a lack of observed blow-up. Positivity, finiteness and reference sensitivity are separate diagnostics; Section 4.4 analyses the price multipliers.

3 Implementation and experimental protocol

3.1 Coupled paths and the role of the grid

A grid is the set of time points where a method updates its states. Finer grids have shorter steps and more updates per path. Let δ be the finest step and $h = r\delta$. Every coarser increment is formed by addition:

$$\Delta W_n^{(h)} = \sum_{j=nr}^{(n+1)r-1} \Delta W_j^{(\delta)}. \tag{27}$$

We aggregate each driver separately, preserving its variance and driver correlation. We then rerun each method's states from the initial conditions on its own grid. We do not merely select a subset of fine-grid prices. Independent paths across grids would measure path mismatch rather than strong discretisation error.

Algorithm 1 Coupled terminal comparison with bounded-memory batches

Input: Model, initial data, T, nested grids, path count M, seed, batch size
Output: Terminal errors, paired intervals, diagnostics and saved arrays

Initialise a fixed-seed generator and streaming statistics
for each independent batch of paths **do**
    Generate Brownian increments on the finest grid
    For the non-affine model, correlate the two drivers by equation (21)
    for each numerical grid **do**
        Sum fine increments using equation (27)
        Reset the state and apply its update rule for all time steps
        Check finite states and collect terminal prices
    end for
    Form absolute pathwise errors and signed payoff differences on paired paths
    Accumulate moments and save the specified terminal arrays
end for
Export CSV/JSON evidence and regenerate figures and report tables

3.2 Sampling intervals, fitted rates and control variates

For M independent observations $A_i = |S_{T,i}^{(h)} - S_{T,i}^{\mathrm{ref}}|$ and $D_i = \varphi(S_{T,i}^{(h)}) - \varphi(S_{T,i}^{\mathrm{ref}})$, use

$$\widehat{e}_s = \overline{A}, \qquad \widehat{b} = \overline{D}, \qquad \widehat{\mathrm{SE}}(\overline{U}) = \frac{s_U}{\sqrt{M}}, \qquad \mathrm{CI}_{95}(\overline{U}) \approx \overline{U} \pm 1.96\frac{s_U}{\sqrt{M}}. \tag{28}$$

Independence between paths gives the standard-error scale; the central limit theorem supplies the approximate normal interval when variance is finite. Sampling intervals are pointwise, not simultaneous, and exclude numerical-reference bias. Signed weak intervals are retained even when they cross zero. Taking the logarithm of an absolute noisy bias does not resolve its sign or justify a rate.

Production fitted rates use explicitly stated windows. Levels share paths, so an ordinary regression covariance matrix would not be an independent-level uncertainty calculation. Additional independent-seed runs in Section 4.7 preserve within-run coupling and assess across-run variability instead.

The GBM mean bias is small relative to raw sampling noise. We use five exactly centered controls C and a coefficient vector β fitted on a separate pilot, then estimate $D - \beta^\top C$ in production. Conditional on the independent pilot,

$$\mathbb{E}[D - \beta^\top C \mid \beta] = \mathbb{E}[D], \qquad \mathbb{E}[C] = 0. \tag{29}$$

This reduces sampling variance without substituting equation (20) for simulated results. Appendix G gives the control definitions; raw and controlled estimates remain in the CSV.

3.3 Files, units and design decisions

Algorithm 1 explains the shared-noise workflow; Table 2 locates its executable components.

Table 2: Executable components in the portable code archive. Paths are relative to its top-level folder.

File	Purpose
`code/gbm.py`	Exact GBM, price EM/Milstein and independent-pilot controls
`code/nonaffine.py`	Simultaneous log-EM, nested grids and state diagnostics
`code/verify.py`	Separate recurrence/quadrature checks and saved-data consistency
`code/build_report_data.py`	Tables, numeric macros and production-report plots
`code/submission_experiments.py`	Independent-seed repeat study and measured kernel-cost comparison
`code/run_all.py`	One-command reproduction; isolated smoke-test mode

All model parameters are named constants or dictionaries; no market-data download is required. A stable logistic evaluation avoids exponentiating a large positive number. Batching controls memory but is part of reproducibility, since changing batch shape can assign pseudo-random draws to different paths. The non-affine Numba kernel accelerates time stepping but does not supply an external exact sampler. The model-specific integrator is intentionally specialised to this assignment; its constant-volatility reduction supplies a swappable special-case verification.

Table 3: Computational protocol; grids are nested powers of two.

Quantity	Value	Quantity	Value
GBM production paths	160,000	Independent pilot	20,000
GBM seed	2026090804	Pilot seed	2026090805
GBM grids N	8 to 1024	GBM batch	2000
Non-affine paths	100,000	Seed	2026090804
Coarse N	8,16,32,64,128,256	Reference N	2048,4096,8192
GBM elapsed	30.9 s	Non-affine elapsed	107.8 s

Table 3 records the production protocol and its measured original run costs. The GBM rate window is $N = 64, 128, 256, 512, 1024$; the non-affine principal strong window is $N = 8, 16, 32, 64$. Choices not specified in the pack include GBM parameters, path counts, seeds, grids, batch sizes, controls, fit windows and the practical 10% reference-sensitivity screen. These choices are recorded, not presented as assignment-supplied constants. The second model's parameters are those in Table 1. The call observable $\varphi(S) = (S - 100)^+$ is an *undiscounted expected payoff* under the supplied drift. It is not a risk-neutral option price without a pricing measure and discount rate.

4 Numerical experiments

4.1 Verification against independently constructed checks

The exact GBM solution in equation (3) is our main pathwise oracle. The verification script also independently recomputes the EM/Milstein recurrences with a loop, uses Gaussian quadrature for multiplier moments, recomputes non-affine simultaneous updates in scalar Python, and reduces the non-affine model to constant volatility. In that special case log-EM must agree with exact GBM on every grid. Finally it recomputes reported strong and signed weak statistics from the saved terminal arrays. These checks test the imported executable functions rather than a disconnected explanatory listing. Recorded verification and smoke-test evidence accompany the archive. Automated checks do not constitute a signed human code review.

4.2 GBM strong convergence

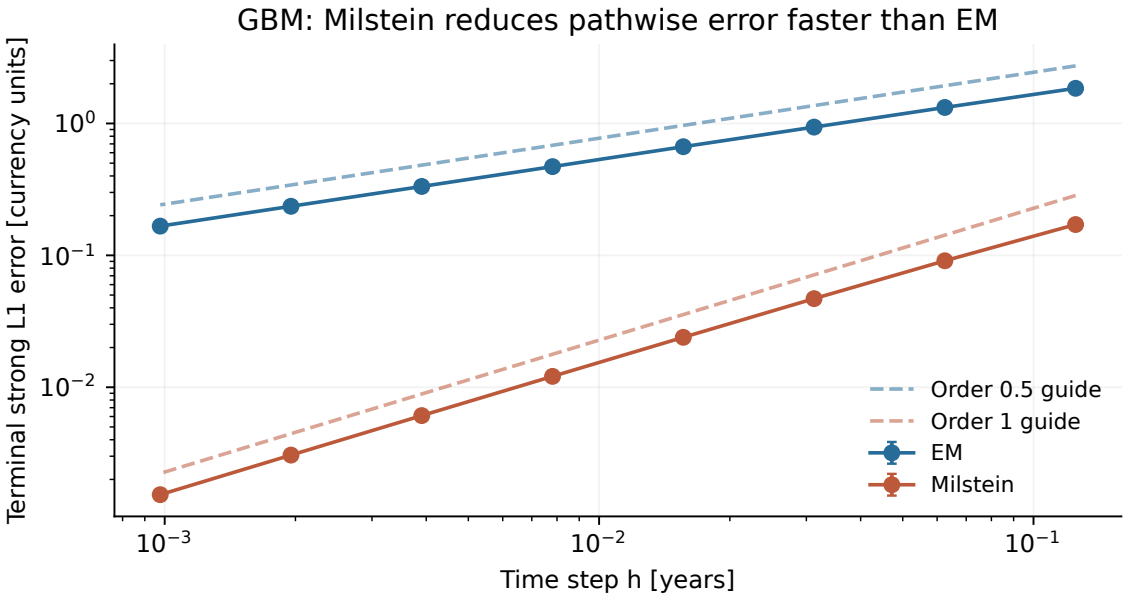


Figure 1: Coupled GBM terminal L^1 error. Reference lines benchmark orders 1/2 and 1; pointwise 95% sampling intervals are often smaller than the markers. Milstein's pathwise error decays faster.

Table 4: GBM strong error estimates with 95% interval half-widths.

h	EM	Milstein
1/8	1.84750 ± 0.00809	0.17127 ± 0.00100
1/16	1.32218 ± 0.00562	0.09096 ± 0.00048
1/32	0.93847 ± 0.00396	0.04697 ± 0.00023
1/64	0.66641 ± 0.00278	0.02391 ± 0.00011
1/128	0.47071 ± 0.00196	0.01209 ± 0.00006
1/256	0.33297 ± 0.00139	0.00610 ± 0.00003
1/512	0.23550 ± 0.00098	0.00306 ± 0.00001
1/1024	0.16658 ± 0.00069	0.00153 ± 0.00001

Figure 1 and Table 4 give production slopes 0.500 for EM and 0.991 for Milstein on $N = 64, 128, 256, 512, 1024$. The distinction agrees with Section 2.2. The comparison uses the same Brownian paths, not separate realizations. Exact simulation has no time-discretisation error at the grid points, although floating-point sums can change when regrouped.

4.3 GBM weak mean convergence

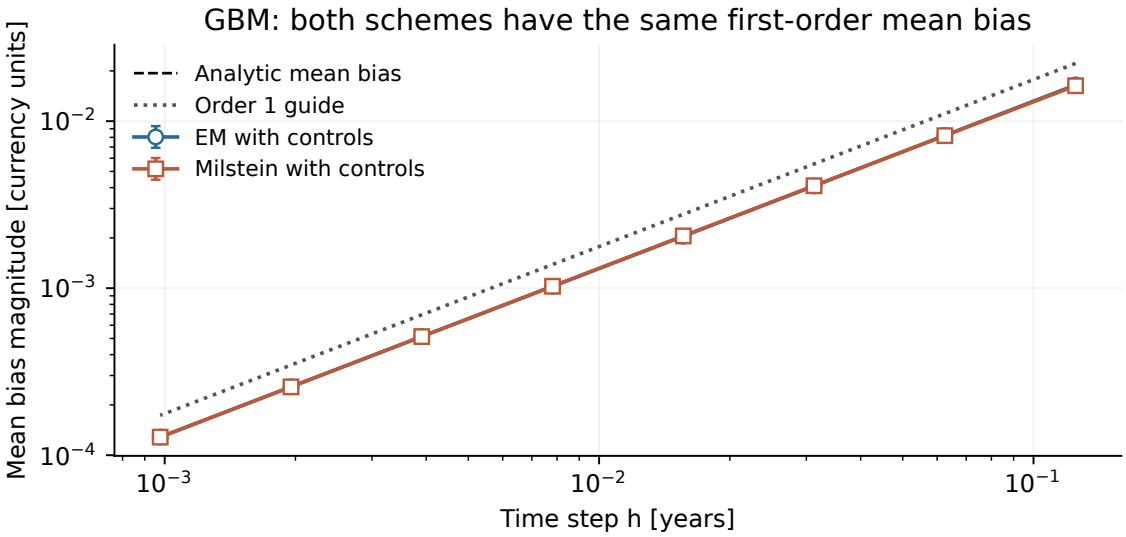


Figure 2: Controlled paired mean-bias estimates and the analytic discrete-mean bias. Both methods approach the same order-one line because their expected one-step multipliers coincide, not because their paths have equal accuracy.

Table 5: Signed weak bias; controlled estimates plus 95% half-widths.

h	Analytic bias	EM controlled	Milstein controlled
1/8	-1.636×10^{-2}	$-1.649 \times 10^{-2} \pm 1.940 \times 10^{-4}$	$-1.629 \times 10^{-2} \pm 9.901 \times 10^{-5}$
1/32	-4.102×10^{-3}	$-4.096 \times 10^{-3} \pm 2.811 \times 10^{-5}$	$-4.106 \times 10^{-3} \pm 1.399 \times 10^{-5}$
1/128	-1.026×10^{-3}	$-1.025 \times 10^{-3} \pm 3.653 \times 10^{-6}$	$-1.027 \times 10^{-3} \pm 1.810 \times 10^{-6}$
1/512	-2.566×10^{-4}	$-2.567 \times 10^{-4} \pm 4.660 \times 10^{-7}$	$-2.566 \times 10^{-4} \pm 2.301 \times 10^{-7}$
1/1024	-1.283×10^{-4}	$-1.284 \times 10^{-4} \pm 1.649 \times 10^{-7}$	$-1.283 \times 10^{-4} \pm 8.144 \times 10^{-8}$

Equation (19) explains the near-overlap in Figure 2. The analytic fitted slope is 0.9993; controlled production slopes are 0.999 and 1.000. At $N = 1024$ the analytic bias magnitude is only about 1.28×10^{-4} currency units. The raw EM standard error is larger, so the archive retains both the unadjusted estimates and the variance-reduced estimates. Their independence from the pilot, rather than agreement with a desired curve alone, protects the interpretation. Table 5 compares actual estimates with the analytic oracle.

4.4 Distribution and positivity

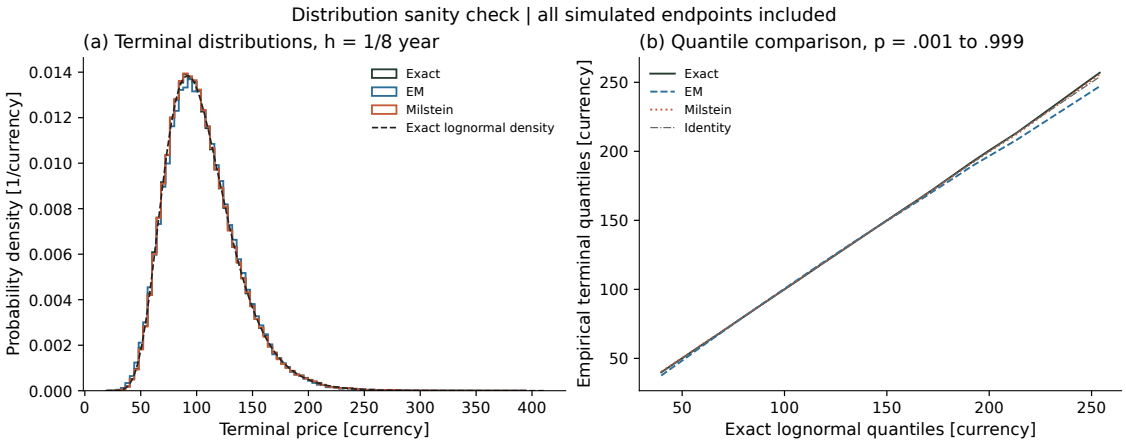


Figure 3: GBM terminal histogram and exact lognormal density. The coarse $h = 1/8$ comparison tests distributional agreement; quantiles in Table 6 add a separate numerical check.

Table 6: Terminal quantiles; numerical columns use $h = 1/8$.

p	Lognormal	Exact simulation	EM	Milstein
0.01	50.012	49.928	48.387	50.213
0.05	61.357	61.378	60.512	61.615
0.25	82.091	82.151	82.303	82.263
0.50	100.501	100.557	101.088	100.582
0.75	123.041	123.164	123.653	123.057
0.95	164.618	164.485	163.370	164.043
0.99	201.961	202.402	198.376	201.539

Figure 3 checks the density and empirical quantiles. Exact-simulation empirical mean and variance are 105.1987 and 1045.0120, and its mean interval is [105.0403, 105.3571]. The theoretical targets are 105.1271 and 1040.7868. Sampling fluctuations remain even when a sampler is exact.

For a positive old price, EM's one-step negative probability and Milstein's completed-square multiplier are

$$\mathbb{P}(S^E_{n+1} < 0 \mid S^E_n > 0) = \Phi\left(-\frac{1+\mu h}{\sigma\sqrt{h}}\right), \tag{30}$$

$$A_M = \tfrac{1}{2}\sigma^2(\Delta W + 1/\sigma)^2 + \tfrac{1}{2} + (\mu - \tfrac{1}{2}\sigma^2)h. \tag{31}$$

For $\sigma > 0$, a negative Milstein multiplier has positive probability exactly when $(\sigma^2 - 2\mu)h > 1$. In this baseline $\sigma^2 - 2\mu = -0.01$, so Milstein is positive for all $h > 0$; this is not a universal statement about Milstein. The saved terminal nonpositive counts are zero out of 160,000 paths for both methods at every $N = 8, 16, 32, 64, 128, 256, 512, 1024$. These counts imply zero observed strictly negative terminals, but do not count paths ever becoming negative between terminals. Zero observations do not prove EM positivity preservation. Negative prices could distort nonlinear payoffs; clipping them would also change and bias the scheme. Neither price scheme is clipped. This optional positivity investigation is limited to formulas and terminal frequencies, not a stress-test pricing study.

4.5 Non-affine strong differences and reference sensitivity

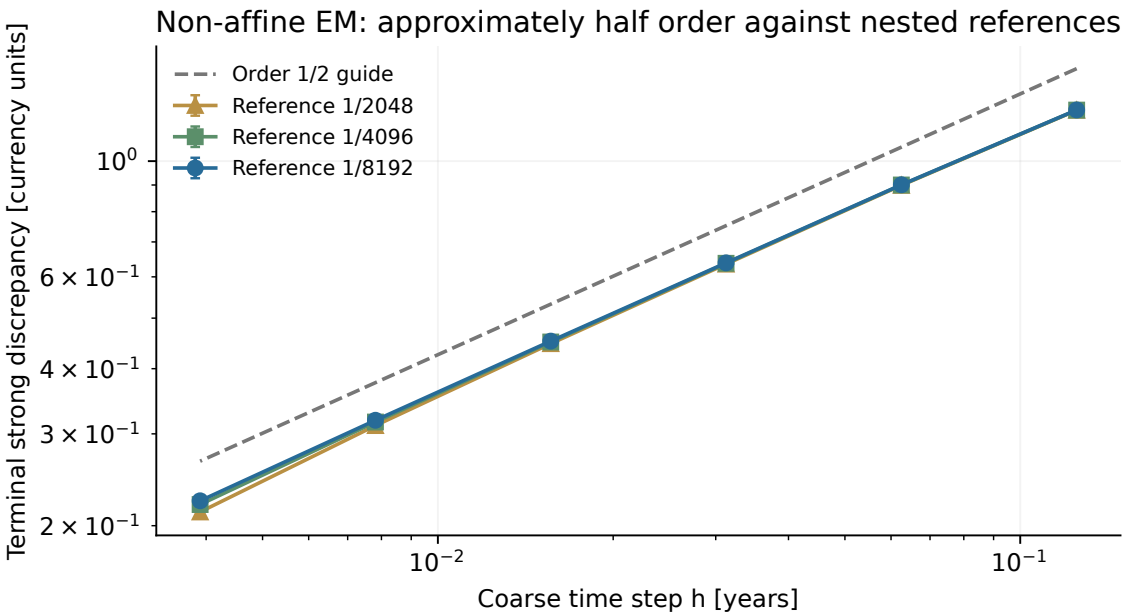


Figure 4: Terminal strong differences against three coupled numerical references. The approximately half-order slope is empirical; agreement of two EM grids is not error against an exact solution.

Table 7: Strong discrepancy against $h_{\text{ref}} = 1/8192$; 95% half-widths.

h	$\mathbb{E}\lvert S_h - S_{\text{ref}}\rvert$	Last reference difference / error
1/8	1.25354 ± 0.00702	3.2%
1/16	0.90087 ± 0.00489	4.5%
1/32	0.63818 ± 0.00341	6.3%
1/64	0.45176 ± 0.00241	8.9%
1/128	0.31854 ± 0.00170	12.6%
1/256	0.22323 ± 0.00119	18.0%

Against $N_{\text{ref}} = 8192$, the empirical slope is 0.491 on $N = 8, 16, 32, 64$. Table 7 records the finest-reference estimates, while Figure 4 displays all three references so the rate is not tied to one unexamined benchmark. The fitted slopes for references 2048, 4096 and 8192 are approximately 0.496, 0.493 and 0.491. Finer coarse points are more sensitive to reference resolution. With $S^\star$ the unknown continuous-time terminal price, the reverse triangle inequality yields

$$\lvert \mathbb{E}\lvert S_h - S^\star\rvert - \mathbb{E}\lvert S_h - S_r\rvert \rvert \le \mathbb{E}\lvert S_r - S^\star\rvert. \tag{32}$$

The unknown right-hand side is not replaced by a reference-to-reference difference. We report the latter only as sensitivity evidence.

Table 8: Coupled refinement of the numerical reference.

Refinement $N_a \to N_b$	Strong difference	Signed payoff difference and 95% CI
2048 → 4096	0.05679 ± 0.00030	-0.000023 [-0.000423, 0.000377]
4096 → 8192	0.04016 ± 0.00021	0.000094 [-0.000188, 0.000376]

The last refinement discrepancy is 0.04016. It is 3.2% to 8.9% of the coarse strong differences on the prespecified fitting window. This meets the practical 10% reference-sensitivity criterion

used for that window. For $N = 256$, the ratio is 18.0%, so the finest coarse points are retained as sensitivity evidence, not used for the principal rate claim. The two refinements decrease the strong discrepancy; their paired payoff intervals assess whether the benchmark changes at a detectable weak scale. Neither a small refinement difference nor a zero-containing interval is a rigorous bound on the unknown continuous-time error. Table 8 supplies the required two further reference refinements. The last strong discrepancy is about 0.04016 currency units, with substantial impact on the finest coarse grids. The 10% screen is a chosen practical diagnostic, not a rigorous exact-error bound.

4.6 Non-affine weak payoff differences

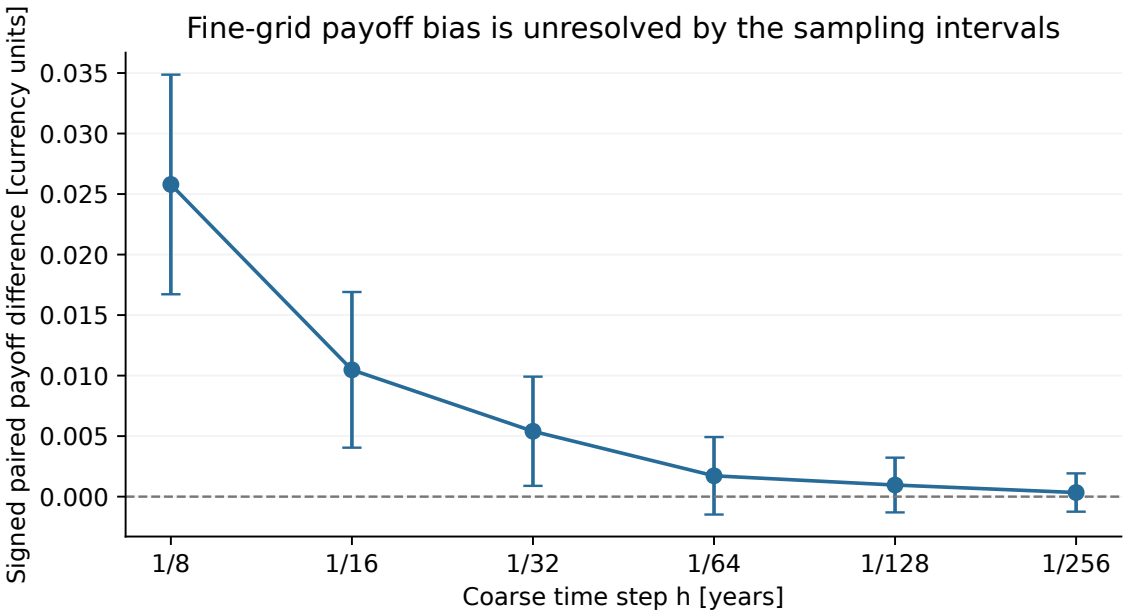


Figure 5: Signed call-payoff differences relative to $N_{\text{ref}} = 8192$. The zero line and paired sampling intervals show why a weak slope is not assigned to the fine-grid estimates. A signed linear ordinate avoids taking logs of intervals crossing zero.

Table 9: Signed paired payoff differences against $h_{\text{ref}} = 1/8192$.

h	Estimated bias	95% interval	Zero excluded
1/8	0.02579	[0.01672, 0.03486]	Yes
1/16	0.01048	[0.00405, 0.01691]	Yes
1/32	0.00540	[0.00089, 0.00991]	Yes
1/64	0.00172	[-0.00148, 0.00492]	No
1/128	0.00096	[-0.00130, 0.00322]	No
1/256	0.00034	[-0.00124, 0.00192]	No

The 95% intervals include zero at $N = 64, 128, 256$. No weak order is assigned to the complete grid range. The finest-reference payoff is estimated as 14.53804 with interval [14.40515, 14.67093]. The payoff has a corner at the strike, so smooth-test-function weak-order results cannot simply be assumed. Figure 5 and Table 9 quantify signed sampling uncertainty, not numerical-reference bias.

For the continuous model, bounded g makes the stochastic exponential a true finite-horizon martingale, so the mean remains $S_0 e^{\mu T}$. The discrete log-EM method preserves that same mean

at any step size:

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_{t_n}] = S_n e^{(\mu - g(Y_n)^2/2)h} \mathbb{E}[e^{g(Y_n)\Delta W_n^{(1)}} \mid \mathcal{F}_{t_n}] = S_n e^{\mu h}. \tag{33}$$

The Gaussian exponential expectation cancels the variance correction. Thus an accurate mean alone cannot test weak time bias for this model. This motivates a nontrivial payoff observable.

Table 10: Observed Brownian and state diagnostics. Increment moments are measured from generated fine increments; state checks include intermediate updates.

Quantity	Observed	Target or scope
Increment variance 1	1.221×10^{-4}	1.221×10^{-4}
Increment variance 2	1.221×10^{-4}	1.221×10^{-4}
Increment correlation	-0.699675	-0.700000
Sampled increments	4,194,304	Predetermined first batch
Total path updates	1,484,000,000	All grids
Finite intermediate states	Pass	X and Y
Positive finite prices	Pass	All observed states

The finest mean is 105.08796 with sampling interval [104.89962, 105.27629], consistent with its exact first-moment target. The expected payoff estimate is 14.53804 with interval [14.40515, 14.67093] for the finest *discrete* model. Table 10 checks both driver variances against h, correlation against -0.7, and finite positive asset states.

4.7 Independent-seed variability of fitted rates

8 independent-seed GBM runs use 5,000 production paths and a separate 1,000-path control pilot each, with the same $N = 64, 128, 256, 512, 1024$ fit window. 8 non-affine runs use 1,000 paths each, the same $N = 8, 16, 32, 64$ window and $N_{\text{ref}} = 8192$. These are additional experiments, not replacements for the 160,000- and 100,000-path production studies.

Table 11: Across-seed fitted-rate variability. Student-t intervals concern the mean fitted slope under this fixed setup, not an exact theorem about the convergence order.

Study	Mean slope	Seed SD	Approx. 95% mean-slope CI
GBM EM strong	0.5003	0.0083	[0.4933, 0.5072]
GBM EM weak	0.9987	0.0061	[0.9936, 1.0037]
GBM Milstein strong	0.9928	0.0032	[0.9902, 0.9955]
GBM Milstein weak	1.0003	0.0029	[0.9979, 1.0028]
Non-affine EM strong	0.4984	0.0174	[0.4838, 0.5129]

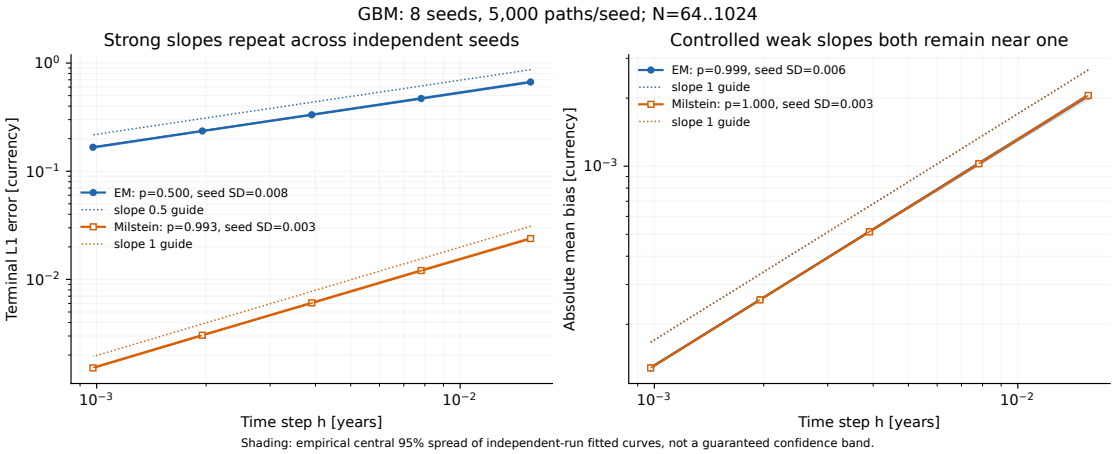


Figure 6: Independent-seed GBM fits with theoretical guides. Shading is the empirical central 95% spread of fitted curves across runs, not an independent-level regression band or a guaranteed confidence band. Strong and weak rates remain distinct.

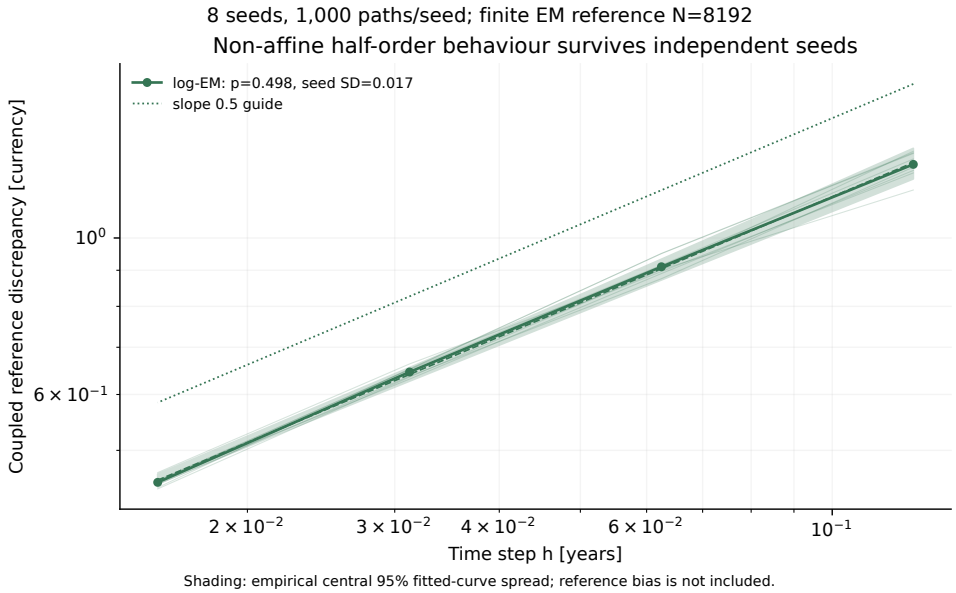


Figure 7: Non-affine strong fits vary across independent seeds but remain near half order for the same numerical reference. This spread does not include the unknown reference bias.

Table 11 and Figures 6–7 show the reproducibility of the fitted-rate conclusions. Only a small number of repeats is used; their empirical spread is descriptive. Coupled levels, finite fit windows and numerical references can produce systematic effects not included in a seed-only interval. In particular, Milstein's mean fitted strong slope is slightly below one and its sampling-only interval excludes one. This reflects a finite-window fitted quantity; it is not a confidence interval for the asymptotic theoretical order.

4.8 Measured cost at a common accuracy threshold

Cost is measured wall-clock latency of the GBM NumPy update kernel, on the same pre-generated coupled Brownian paths after warm-up (10,000 paths; median of 7 timings). The common target is terminal L^1 error at most 0.25 currency units. We select the coarsest tested grid whose upper pointwise 95% error interval is below this target; the actual errors need not be identical, and the tolerance overshoot is reported.

Table 12: GBM kernel costs at the common accuracy target. Both methods satisfy the same threshold; their measured errors and intervals are shown rather than assumed equal.

Method	N	Error	95% error CI	Median cost [ms]
EM	512	0.23566	[0.23177, 0.23956]	8.532
Milstein	6	0.21605	[0.21087, 0.22123]	0.139

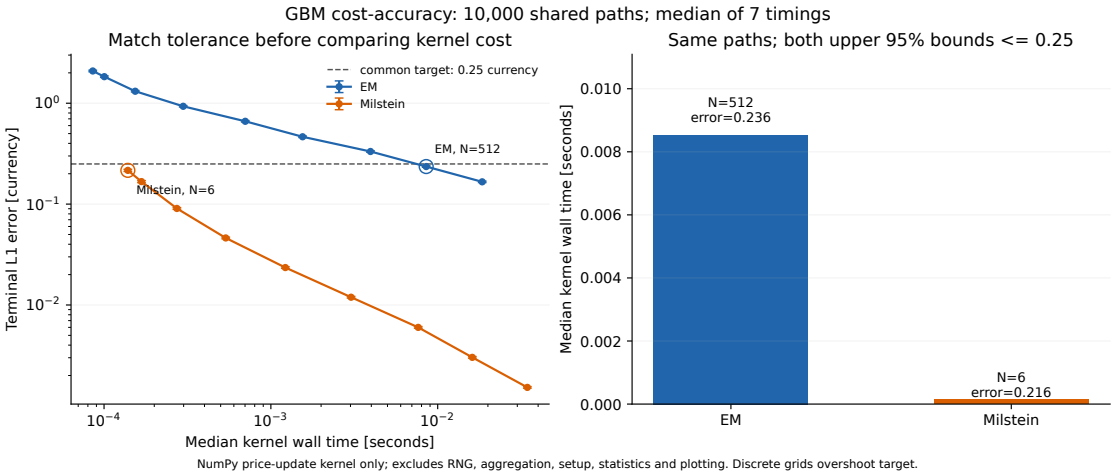


Figure 8: Measured kernel cost versus terminal strong error, using the same accuracy target and coupled paths. The target line prevents interpreting a cheap but inaccurate grid as a successful comparison.

Temporary allocations inside the price-update kernel are counted. Random-number generation, Brownian aggregation, input preparation, import/setup, compilation, statistical summaries, pilot fitting, verification and plotting are excluded from this kernel-only measure and treated consistently across methods. Raw repeated timings and the tested environment are recorded in `results/submission_experiments.json`. The comparison in Table 12 and Figure 8 therefore supports a specific implementation-level cost–accuracy conclusion, not an end-to-end runtime or cross-machine speedup.

5 Conclusions and limitations

GBM verifies the expected strong advantage of Milstein, but both price methods have exactly the same first-order terminal-mean bias. The observed overlap of weak curves has a mathematical explanation; it is not a plotting mistake. Exact simulation is preferable for GBM itself, while the numerical price methods provide a controlled validation exercise.

For the prescribed non-affine model, log-EM preserves positive prices and its first moment. Nested references yield approximately half-order strong differences on the selected coarse window. They do not supply an exact error. Fine-grid payoff differences are unresolved by the pointwise sampling intervals, so no complete-grid weak order is claimed.

Accuracy and work have different components. Halving the time step roughly doubles path updates; four times as many independent paths roughly halves a standard error. Coupling and controls reduce sampling noise, while refinement addresses discretisation. More paths cannot repair an underresolved reference. The supplemental cost evidence is specific to the measured implementation and stated common accuracy threshold, not a portable claim that one method is always faster.

The scope is limited to the stated parameters, horizon, terminal L^1 norm, path counts and fit windows. Positivity frequency diagnostics cover terminals for GBM, and all observed states for non-affine EM. We do not claim exact weak payoff errors, unconditional GBM positivity, or a scalar-Milstein rate for a noncommutative two-noise system. A useful next step is a larger paired payoff experiment with stronger reference resolution and variance reduction, followed by stress tests in higher-volatility regimes.

Reproducibility and intellectual ownership

The code archive includes fixed-seed source, saved arrays, numeric results, figure-generation scripts and machine/environment evidence. It supports full reproduction and isolated reduced smoke tests. AI assistance included derivations, explanations, implementation, numerical execution and document preparation. Appendix A records the significant claim audits and distinguishes automated verification from the team's responsibilities. Appendices B–F record the members' individual contributions.

References

[1] *Problem Pack: Five Suggested Directions for the ODE IVP Project.* Supplied coursework document, introductory SDE exception and Direction 4, pp. 6–7.

[2] D. J. Higham. An algorithmic introduction to numerical simulation of stochastic differential equations. *SIAM Review*, 43(3):525–546, 2001. doi:10.1137/S0036144500378302.

[3] P. E. Kloeden and E. Platen. *Numerical Solution of Stochastic Differential Equations.* Springer, 1992. doi:10.1007/978-3-662-12616-5.

[4] C. R. Harris et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi:10.1038/s41586-020-2649-2.

[5] S. K. Lam, A. Pitrou and S. Seibert. Numba: A LLVM-based Python JIT compiler. *LLVM-HPC*, 2015. doi:10.1145/2833157.2833162.

A Team AI Transparency Log

This log records the AI assistance used in the project, checks of significant scientific claims, and the team's intellectual-ownership responsibilities.

A1. AI tools used

Table 13 summarises the tools and their uses.

Table 13: AI assistance used during the project.

Tool	Version/date	Primary use
OpenAI Codex	September–October 2026; exact model versions were not recorded.	Derivations and explanations; implementation and automated verification; experiment execution; report, figures, presentation and documentation preparation.

AI output was treated as material to check and interpret, not as an independent authority. Mathematical and implementation assistance are disclosed alongside language and formatting assistance. Raw conversation transcripts are not appended; no transcript archive or content hash is claimed.

A2. Significant AI-assisted claim audits

Table 14: Scientific claims, their assumptions and ways they could fail.

Item	Claim, assumptions and material question	Risk and possible falsifier
1	Explain why the GBM EM and Milstein weak-mean curves overlap. Claim: both discrete means equal the expression in equation (19) for deterministic initial data and independent centered Gaussian increments.	Confusing strong with weak order, omitting the centering of the Milstein term, or presenting the analytic curve as Monte Carlo output. A multiplier expectation different from $1 + \mu h$ falsifies the identity.
2	Implement and interpret a numerical reference for the non-affine model: sum shared fine increments, rerun simultaneous old-Y EM, and compare three reference grids.	Updating X with new Y, using independent paths across resolutions, or calling grid differences exact errors. A constant-volatility reduction failing the exact GBM test or large reference sensitivity falsifies the implementation/interpretation.
3	Interpret empirical slopes and cost. Claim: Milstein improves strong GBM accuracy but not this mean bias; finer-grid non-affine payoff slopes are unresolved.	Treating correlated fit residuals as independent uncertainty; overlooking zero-containing payoff intervals; comparing speed at different accuracy requirements. Independent-seed spread, signed paired intervals, or matched-target cost can contradict the stated interpretation.

A3. Verification, decision and limitations

Table 15: Checks supporting the claim audits in Table 14.

Item	Separate numerical check and evidence	Decision, modification and limitation
1	A direct conditional-expectation calculation in equations (17)–(20); separate Gaussian-quadrature checks of discrete multiplier moments in `code/verify.py`; raw and controlled columns in `results/gbm_convergence.csv`.	Retain both curves. Explain their identical first-order mean bias rather than label the overlap a plot error. The statement applies to $\varphi(S) = S$ and $\mu \neq 0$, not every payoff.
2	Scalar Python recurrence and constant-volatility oracle in `code/verify.py`; saved-array recomputation; three-grid refinements in `results/nonaffine_reference_checks.csv`.	Keep log-EM and simultaneous updates. Call strong rates empirical. Treat the 10% screen as a chosen sensitivity diagnostic, not a rigorous residual bound; do not add an unjustified scalar Milstein correction.
3	Eight independent-seed repetitions and kernel-cost measurement in `code/submission_experiments.py`; recorded JSON/CSV and plots; pointwise production payoff intervals.	Add across-run rate variability and same-target cost evidence. Withhold a fine-grid weak-order claim. Seed spread excludes numerical-reference and fit-window bias; kernel timing excludes random-number generation and other costs.

The checks in Table 15 and the separate implementations were produced with AI assistance; they are not presented as an independent human audit. The team reviewed the formulas, implementation interfaces and interpretation according to the responsibilities in Table 16 and the individual statements. Passing automated tests is distinguished from understanding or proving a mathematical claim.

A4. Intellectual ownership declaration

We affirm responsibility for the team-owned methods and claims described in this report. We can explain our contributions, justify the stated conclusions and limitations, and describe or modify the relevant methods without AI assistance during the viva. Trusted library operations are distinguished from the mathematical model and numerical choices owned by the team.

Table 16: Individual responsibilities and trusted interfaces. The two lines in each signature cell are for signature and date, respectively.

Name	Signature / date	Team-owned contribution	Trusted dependency/interface
Yifan Chen	_____ _____	Reviewed the models, log-price transformation, EM/Milstein formulas and strong–weak convergence interpretation.	Gaussian increment moments and scalar SDE calculus underpin the derivations; numerical library checks support, but do not replace, those calculations.
Zeya Hu	_____ _____	Checked GBM coupling, update multipliers, independent-pilot controls and moment/distribution diagnostics.	NumPy normal sampling, array aggregation and products; their shapes, scaling and random-seed protocol are specified by the project.
Yanle Li	_____ _____	Checked correlated increments, simultaneous log-EM, nested-grid construction and reference-sensitivity interpretation.	NumPy sampling and aggregation; Numba compilation of the EM kernel, with the recurrence and old-state conventions owned by the team.
Qingwen Ma	_____ _____	Reviewed figures, signed confidence intervals, independent-seed rate variability and common-accuracy kernel costs.	NumPy statistical summaries, Matplotlib plotting and Python timing; the error metric, fit windows and timing boundary are explicit project choices.
Chenyi Xu	_____ _____	Checked report consistency, reproduction commands, test/evidence distinctions and report/code archive organisation.	Python filesystem and ZIP interfaces and LaTeX typesetting; package paths, dependency instructions and document content are checked by the team.

B Individual Contribution Statement: Yifan Chen

Name: Yifan Chen **Student ID:** 2362353
Role this project: Mathematical models, numerical methods and convergence interpretation.

The specific things I did this project.

1. I reviewed the GBM SDE, the Itô transformation to log price and the exact terminal moments in Section 1.2. I checked that log-price EM is exact for this constant-coefficient model, whereas the convergence comparison uses price-based EM and Milstein.
2. I checked the conditional-expectation calculation in equations (17)–(20). I clarified why the two methods have identical terminal-mean bias without having identical pathwise error.
3. I reviewed the strong–weak distinction and local-expansion discussion in Section 2.2, together with the non-affine commutator in equation (25). I checked the limitation on transferring scalar Milstein order to the two-driver model.

The hardest conceptual obstacle and how I addressed it.

The main issue was reconciling Milstein's better strong accuracy with the overlapping weak-mean curves. I separated the pathwise error from the selected observable and checked the expectation of each update multiplier. This explained the overlap without treating it as a plotting defect.

An AI-assisted output I used, what I changed and why.

I reviewed the AI-assisted weak-mean explanation recorded in Appendix A, audit item 1. I retained the discrete-mean identity but restricted its interpretation to the mean observable; it does not imply equal accuracy for every payoff or path.

What I would do differently.

I would define the observable and error metric before discussing convergence orders. I would also place the update formula and its expectation side by side for easier checking.

One thing I still do not fully understand.

The full construction and simulation of multidimensional Milstein iterated integrals remains beyond my verified contribution. The report therefore does not claim that method's order for the non-affine model.

Anything I should have asked for help with earlier.

A lesson for future work is to agree on the strong and weak error definitions before interpreting convergence plots.

Signature: _____ **Date:** _____

C Individual Contribution Statement: Zeya Hu

Name: Zeya Hu **Student ID:** 2362375
Role this project: GBM implementation, variance reduction and numerical checks.

The specific things I did this project.

1. I reviewed `code/gbm.py`, particularly `simulate_batch`, against equations (12) and (13). I checked that the resolutions aggregate common fine increments and compare their terminal prices with the same exact GBM path.
2. I checked `fit_controls` and the independent pilot protocol in Section 3. I distinguished the raw Monte Carlo estimates, controlled estimates and analytic bias in `results/gbm_convergence.csv`.
3. I reviewed `exact_moments` and `analytic_discrete_moments`, the separate GBM checks in `code/verify.py`, and the distribution and positivity evidence in Section 4.4.

The hardest conceptual obstacle and how I addressed it.

The main issue was making a small mean bias visible despite sampling noise. I examined the paired comparison and the independent fitting of control coefficients. I checked that variance reduction changes the sampling variability rather than replacing the numerical mean by its theoretical value.

An AI-assisted output I used, what I changed and why.

I reviewed the AI-assisted GBM implementation and interpretation associated with Appendix A, audit items 1 and 3. I checked the code against the update formulas and preserved the distinction between simulated estimates and the analytic reference curve.

What I would do differently.

I would document the raw estimator before introducing control variates. I would add tests using several drift and volatility settings, rather than relying only on the production baseline.

One thing I still do not fully understand.

I have not established how efficient the selected controls remain for different parameters or nonlinear payoffs. Their observed performance here is not a general variance-reduction guarantee.

Anything I should have asked for help with earlier.

For future work, I would discuss the anticipated bias and sampling precision before fixing the path count and control design.

Signature: _____ **Date:** _____

D Individual Contribution Statement: Yanle Li

Name: Yanle Li **Student ID:** 2363600
Role this project: Non-affine model, correlated increments and numerical-reference validation.

The specific things I did this project.

1. I reviewed `brownian_increments` in `code/nonaffine.py` against equation (21). I checked the increment variance, prescribed correlation and construction from two independent normal variables.
2. I checked `_em_kernel` against equations (22)–(24). I verified the simultaneous use of the old volatility factor and reviewed the log-price positivity and finiteness diagnostics.
3. I examined `em_terminal`, its nested-grid checks and increment aggregation. I reviewed the three-reference comparisons in Section 4.5 and `results/nonaffine_reference_checks.csv`, distinguishing numerical differences from exact errors.

The hardest conceptual obstacle and how I addressed it.

The main issue was understanding what a numerical reference can establish when an exact transition sampler is unavailable. I followed the shared increments through each grid and separated reference-refinement differences from an unknown residual error against the exact solution.

An AI-assisted output I used, what I changed and why.

I reviewed the AI-assisted reference implementation described in Appendix A, audit item 2. I checked aggregation and old-factor updates, and retained the qualification that the fitted strong rate is empirical and the reference-sensitivity screen is not a rigorous bound.

What I would do differently.

I would illustrate the aggregation of increments with a small explicit grid example before presenting the production protocol. Additional fine references would strengthen the sensitivity study when computational resources permit.

One thing I still do not fully understand.

A certified bound for the exact-solution residual of the finest reference is not established here. Agreement between two fine EM grids alone does not supply that bound.

Anything I should have asked for help with earlier.

A useful future checkpoint is to agree on the meaning and limitations of a reference solution before choosing the fitting window.

Signature: _____ **Date:** _____

E Individual Contribution Statement: Qingwen Ma

Name: Qingwen Ma **Student ID:** 2361052
Role this project: Statistical interpretation, visualisation and cost–accuracy comparison.

The specific things I did this project.

1. I checked the strong and weak plots against their saved CSV results and captions in Section 4. I reviewed axis scales, theoretical reference lines and signed paired confidence intervals, particularly Figure 5.
2. I reviewed the functions `gbm_repeat_study` and `nonaffine_repeat_study` in the supplementary experiment script, `code/submission_experiments.py`. I checked the distinction between independent-seed variability and errors caused by the reference grid or fitting-window choice.
3. I examined `cost_accuracy_study` and Section 4.8. I checked that methods are compared at a common accuracy target and that the reported kernel timings exclude random-number generation and other workflow costs.

The hardest conceptual obstacle and how I addressed it.

The main issue was deciding whether small weak differences support a convergence slope. I checked the signed intervals before considering a logarithmic fit. Intervals crossing zero do not resolve the bias magnitude, so I retained the report's decision not to assign a fine-grid weak order.

An AI-assisted output I used, what I changed and why.

I reviewed the AI-assisted slope and cost interpretation in Appendix A, audit item 3. I kept the uncertainty qualifications and distinguished across-run slope spread from a confidence statement about exact-solution error.

What I would do differently.

I would set the required precision before selecting the weak-error sample size. I would complement kernel timings with end-to-end measurements, using the same accuracy criterion for each method.

One thing I still do not fully understand.

A combined uncertainty assessment incorporating sampling, fitting-window choice and numerical-reference bias remains unresolved. The reported intervals and repeated-seed spread measure only specified components.

Anything I should have asked for help with earlier.

For future work, I would agree on the interpretation of zero-containing intervals and the cost boundary before producing comparison figures.

Signature: _____ **Date:** _____

F Individual Contribution Statement: Chenyi Xu

Name: Chenyi Xu **Student ID:** 2359968
Role this project: Report consistency, reproduction checks and submission organisation.

The specific things I did this project.

1. I reviewed the report against Problem 4 and the supplied submission materials. I checked the connection between parameters, experiment descriptions, figure captions and saved numerical results, without treating the generic ODE tasks as additional SDE requirements.
2. I examined `check_saved_outputs` in `code/verify.py` and the checks in `tests/test_numerical_methods.py`. I distinguished the small smoke workflow, automated verification and recorded production results rather than treating them as interchangeable evidence.
3. I checked the README reproduction commands and the organisation of the code archive through `code/run_all.py` and `code/prepare_submission.py`. I organised the report and code deliverables; the completed presentation is retained as an archived artefact, not a continuing deliverable.

The hardest conceptual obstacle and how I addressed it.

The main issue was separating technical reproducibility from personal intellectual ownership. I checked which claims are supported by formulas, saved results or automated tests, and kept individual understanding and signatures as separate responsibilities.

An AI-assisted output I used, what I changed and why.

I reviewed the AI-assisted report, verification and submission preparation disclosed in Appendix A. I retained disclosure of mathematical and implementation assistance rather than describing it as language editing alone, and checked consistency between the report and archive.

What I would do differently.

I would maintain a requirement-to-evidence index throughout the project. I would also check portable commands and document assembly before the final submission stage.

One thing I still do not fully understand.

Passing the available tests does not establish coverage of all parameter choices or platforms. Behaviour outside the documented protocol requires additional checks.

Anything I should have asked for help with earlier.

A lesson for future work is to agree on document ownership, validation responsibilities and evidence locations at the start of the project.

Signature: _____ **Date:** _____

G Control-variate definitions and reproduction

Let $W = \sum_n \Delta W_n$, $Q = \sum_n [(\Delta W_n)^2 - h]$, $R = \sum_n [(\Delta W_n)^3 - 3h\Delta W_n]$, and $L = e^{\sigma W - \sigma^2 T/2}$. The implemented controls are

$$\begin{aligned} C_1 &= L - 1, & C_2 &= L(W - \sigma T)/\sqrt{T}, \\ C_3 &= L(Q - \sigma^2 T h)/\sqrt{2Th}, & C_4 &= L(R - \sigma^3 T h^2)/\sqrt{6Th^2}, \\ C_5 &= \frac{L[Q^2 - (2Th + 4\sigma^2 T h^2 + \sigma^4 T^2 h^2)]}{\sqrt{8}\, Th}. \end{aligned} \tag{34}$$

Gaussian exponential tilting shifts each increment's mean to σh and preserves its variance h, giving zero expectation for all five controls. Coefficients are fitted only on independent pilot paths. The analytic discrete bias remains a validation target, never a replacement for a production estimate.

```
python -m pip install -r requirements.txt
python code/verify.py
python -m unittest discover -s tests -v
python code/run_all.py --smoke
python code/run_all.py
```

Listing 1: Commands from the extracted code archive's top-level directory.

Full reproduction regenerates production outputs and supplemental evidence. A smoke run is deliberately too small for scientific slope claims and writes to an isolated directory. The README records exact commands, dependency ranges, historical production versions and which cost components are excluded. The supplied report PDF does not require a local LaTeX installation merely to read it.